Two Networks of Cooperation: Which Ukrainian Hromadas Have EU Twin Cities and Which Have Inter-Municipal Cooperation Agreements

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Data and code: Hromada Strategy Collaboration Mapping, DOI 10.5281/zenodo.22552878 (CC BY 4.0)

Abstract

Ukrainian territorial communities (hromadas) cooperate through at least two channels that policy discussions often treat as one: twinning with municipalities in the European Union, and inter-municipal cooperation (IMC) agreements with domestic neighbours under Law 1508-VII. Joining open registry data for all 1,463 hromadas on the KATOTTG administrative code, we find that 326 hromadas have a registry-confirmed EU partner, 415 were party to an IMC agreement active on 24 February 2022, and only 111 are in both groups ($\varphi = 0.07$). Twinning rises steeply with population (2.6% of hromadas under 5,000 inhabitants; 70.5% of those above 50,000), whereas IMC participation does not (31.7% and 25.3%). In logistic models with oblast fixed effects, population has opposite effects on the two outcomes (odds ratio per tenfold increase: 46.8 for twinning, 0.46 for IMC), while own income per capita and participation in donor programmes raise the odds of both. Controlling for these factors, twinning is at most a weak predictor of IMC (OR 1.36, 95% CI 0.95–1.95). Support instruments designed for one network therefore cannot be assumed to reach communities active in the other. Limitations include a pre-war IMC snapshot, incomplete twinning registries and keyword-based theme classification.

Keywords: inter-municipal cooperation; municipal twinning; Ukraine; decentralisation; hromada; open data

1. Introduction

Since the amalgamation reform that began in 2015, Ukraine's local self-government has been organised around roughly 1,450 territorial communities (hromadas). Two forms of cooperation between hromadas and other municipalities are especially visible in donor programming. The first is *inter-municipal cooperation* (IMC) between Ukrainian communities, regulated by the Law "On Cooperation of Territorial Communities" (1508-VII), which provides five legal forms — among them delegation of tasks, joint municipal enterprises, joint projects and joint financing of institutions — and a state registry of agreements. The second is *twinning* and solidarity partnerships with municipalities abroad, which surged after February 2022, supported by networks such as the German Service Agency Communities in One World (SKEW) and by donor matchmaking.

Policy documents frequently refer to both under the umbrella of "municipal cooperation" [1, 2]. Yet the two differ in partners, resources and purpose: a shared administrative-services centre between two villages and a partnership with a German city that delivers generators and training are different activities, probably available to different kinds of communities. A survey of 546 hromadas by USAID DOBRE found that the hardest stage of domestic IMC is partner search and communication (30% of respondents) rather than legal form [3], suggesting that how communities find partners may shape who participates.

This paper asks two descriptive questions. (1) Which hromadas take part in each network? (2) Do the networks overlap beyond what shared community characteristics would predict? We answer them by

joining registry data on both networks with community-level covariates through a common administrative key.

2. Data and methods

2.1 Unit of analysis and key

The universe is the 1,463 hromadas of the open dataset *Hromada Strategy Collaboration Mapping* (government-controlled territory plus hromadas listed as occupied). All joins use the KATOTTG code, which is unique across the universe; hromada names are not unique and were not used as keys.

2.2 Twinning

A hromada is classified as *twinned* if it has at least one partner with confidence level "registry" in the dataset's twinning layer. Registry sources are the SKEW list of German–Ukrainian municipal partnerships (110 hromadas), the AFCCRE directory of French *jumelages* (44) and the partnership map of the Ukrainian ministry on decentralization.ua (282); hromadas may appear in several sources. Weaker evidence (news items, mentions in hromada strategies) accounts for 27 of 1,371 partner links and is excluded.

2.3 Inter-municipal cooperation

A hromada is classified as *IMC-active* if it is a party to an agreement in the Ukrainian registry of cooperation agreements (data.gov.ua, file dated 28 December 2022) that was in force on 24 February 2022. This yields 306 agreements and 415 hromadas. The mapping from agreements to hromadas is taken from the KSE-Loc-Data-Hub network table [4]. The flag is therefore a pre-invasion snapshot, whereas twinning data include partnerships formed later.

2.4 Covariates

Population (2022) and own budget income per capita come from KSE data [4]. Digital-participation level is derived from the e-dem.ua platform statistics (tiers: high, medium, low, not connected). Participation in donor or technical assistance programmes (DOBRE, DECIDE, GIZ, U-LEAD, EGAP, DESPRO, UNDP, among others) is recorded in the dataset's `DonorsPrograms` field and is a lower bound on actual participation.

2.5 Analysis

We report shares with 95% Wilson intervals across population bands, settlement type, oblast, income quartile, e-dem tier and donor participation, and the φ coefficient for overlap, overall and within population bands. We then fit logistic regressions for (a) twinning, (b) IMC and (c) IMC with twinning as an additional predictor. Predictors are $\log_{10}$ population, $\log_{10}$ own income per capita, e-dem tier (reference: not connected), any donor programme and oblast fixed effects; 1,415 hromadas with known population and income enter the models. Odds ratios for population and income refer to a tenfold increase.

For topics, we classified the titles and forms of 899 registry IMC agreements and the 196 foreign agreements in the Law 3668-IX register [5] with one keyword vocabulary, allowing multiple topics per agreement. Because the 3668-IX register records formal international agreements rather than twinning as such, this comparison is indicative only.

All numbers are produced by a single script (Section 7).

3. Results

3.1 Who takes part

Overall, 326 hromadas (22.3%) are twinned and 415 (28.4%) are IMC-active. Table 1 shows participation by population band.

Table 1. Share of hromadas with twinning and IMC by population (95% Wilson CI).

Population	Hromadas	Twinned	IMC-active
< 5,000	265	2.6% (1.3–5.4)	31.7% (26.4–37.5)
5,000–10,000	417	9.8% (7.3–13.1)	29.0% (24.9–33.5)
10,000–20,000	383	20.1% (16.4–24.4)	27.7% (23.4–32.4)
20,000–50,000	299	44.8% (39.3–50.5)	26.8% (22.1–32.0)
> 50,000	95	70.5% (60.7–78.8)	25.3% (17.6–34.8)

(Four hromadas with unknown population are omitted.) Twinning rises monotonically with size; IMC participation is nearly flat and slightly higher in the smallest communities. By settlement type, 51% of urban, 15% of settlement and 9% of rural hromadas are twinned, versus 28%, 22% and 33% for IMC.

Geography also differs. Twinning is concentrated in the west — Lviv (62%), Zakarpattia (58%) and Volyn (50%) oblasts — while IMC participation is highest in Rivne (59%), where only 6% of hromadas are twinned, followed by Lviv (58%), Poltava (57%) and Ivano-Frankivsk (47%). Kyiv oblast has 32% twinned and 14% IMC-active hromadas.

3.2 Overlap

Of 415 IMC-active hromadas, 111 (26.7%) are also twinned; 215 are twinned only, 304 IMC-active only, and 833 in neither group. The φ coefficient is 0.07. Overlap is not merely a size effect: within population bands φ ranges from 0.04 to 0.13, a small positive association.

3.3 Correlates of participation

Table 2. Odds ratios (95% CI) from logistic regressions with oblast fixed effects (n = 1,415).

Predictor	Twinning	IMC
Population, ×10	46.8 (24.5–89.4)	0.46 (0.31–0.66)
Own income per capita, ×10	2.80 (1.51–5.19)	2.02 (1.29–3.17)
Any donor programme	1.75 (1.09–2.81)	1.90 (1.30–2.78)
e-dem "high" vs not connected	1.77 (1.05–2.98)	1.39 (0.88–2.21)

Population acts in opposite directions: conditional on the other covariates and oblast, larger hromadas are far more likely to be twinned and somewhat less likely to be IMC-active. Income and donor participation raise the odds in both networks by similar amounts (confidence intervals overlap). High digital participation is associated with twinning ($p = 0.03$) but not significantly with IMC. When twinning is added to the IMC model, its odds ratio is 1.36 (0.95–1.95; $p = 0.09$).

Donor-programme patterns differ by programme, though subsamples are small. Of 20 hromadas tagged with GIZ, 16 (80%) have a German partner in SKEW, compared with 7.5% of all hromadas; GIZ is

consistent with its role as operator of the German partnership scheme. DOBRE hromadas are more often IMC-active (40%) than twinned (24%), as are U-LEAD hromadas (81% and 56%).

3.4 Topics

Table 3. Most frequent topics (share of agreements; multi-label).

Domestic IMC registry (n = 899; 37% unclassified)		**Foreign agreements, Law 3668-IX (n = 196; 14% unclassified)**	
Roads and infrastructure	21%	Culture	72%
Education	14%	Education	55%
Administrative-services centres	11%	Economy	53%
Health	9%	Tourism	47%
Social services	6%	Governance	32%
Waste	5%	Roads	26%
Fire protection	4%	Cross-border cooperation	22%

Domestic agreements are dominated by service delivery; foreign agreements by broad cultural, educational and economic exchange. The latter frequently use generic wording ("trade-economic, scientific-technical and cultural cooperation"), which assigns several topics to one agreement and inflates their shares. Energy appears in under 1% of both registers, although it is frequently cited as a priority of partnership programmes; plausibly, it is realised through projects and humanitarian assistance rather than signed agreements.

4. Discussion

The two networks recruit different communities. Twinning behaves like a resource that scales with the size and administrative capacity of a municipality; IMC is consistent with a response to local needs that is, if anything, at least as common where communities are small and may be unable to provide services alone. That donor participation and income matter for both networks suggests that some general capacity to cooperate exists, but its effect on the odds is similar and modest, and being in one network adds little to the odds of being in the other once these factors are controlled.

Three implications follow, each as a hypothesis rather than a finding. First, programmes that fund international partnerships and programmes that fund inter-municipal cooperation reach largely different communities; coordination should not assume that the former builds capacity for the latter. Second, small rural hromadas are well represented in IMC but rarely appear in twinning registries, so any new instrument for connecting them with EU peers faces a distinct starting point. Third, the strong western concentration of twinning, and the Rivne–Poltava concentration of IMC, indicate regional ecosystems — donors, facilitators and associations — worth studying directly.

DOBRE's finding that partner search is the main obstacle to IMC [3] fits with these observations: twinning has dedicated matchmaking mechanisms (SKEW and similar schemes), whereas domestic IMC has no comparable national scheme. We did not test whether differences in partner search explain these patterns.

5. Limitations

1. **Timing.** The IMC flag is a pre-invasion snapshot; twinning includes partnerships formed after 2022. This may amplify the difference by size. A current public IMC registry would be needed to check.
2. **Incomplete twinning registries.** SKEW covers Germany, AFCCRE France; other countries come from the ministry's map. True twinning is more extensive and coverage varies by country.
3. **Association, not causation.** The models adjust for observed characteristics only. Interpretations such as international-relations capacity in large municipalities are conjectures.
4. **Topics.** Classification uses keywords on free text; a large share of IMC agreements (generic joint projects and enterprises) remains unclassified, and the 3668-IX register differs from twinning.
5. **Measurement.** Donor participation is a lower bound; population for some occupied hromadas is a 2022 estimate and unknown for four.
6. **Single snapshot.** No temporal dynamics are examined.

6. Conclusion

In Ukraine, EU twinning and domestic inter-municipal cooperation are largely separate networks recruiting different kinds of communities. Size is the main dividing line, with opposite effects; income and donor engagement favour both. Treating them as one phenomenon in policy design, evaluation or support programmes risks missing most of the communities in the other network.

7. Data and code availability

All inputs except two re-fetchable caches are released in `data/releases/` (CC BY 4.0; dataset card: `DATASET.md`; CSV mirrors in `csv/`). The IMC registry (`yarn fetch-mss-registry`) and the KSE network table [4] are cached under `data/cache/` and are not redistributed. The script `scripts/analysis/twinning_vs_mms.py` reproduces every number in this paper and writes `internal/twinning-vs-mms-results.json`; the regression block requires `pandas` and `statsmodels`.

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